Dr. Peet Cremer

* 27/01/1988

Transistorfaret 9, 1396 Billingstad, Norway | +47 917 42 339

peet.cremer@gmail.com | Peet Cremer | @PeetCremer | Publication list



Senior Principal AI Researcher building time series foundation models for industrial applications. Built an AI research lab from zero, combining hands-on model development with research direction and team leadership.
Ph.D. physicist with a track record of turning applied ML research into production systems and advancing an AI-first, agent-enhanced development lifecycle.

RESEARCH & TECHNICAL ACHIEVEMENTS

- Built Cognite's time series AI research lab from zero and led a team of four to develop a state-of-the-art industrial time series foundation model in six months (April–September 2026); benchmarks reported at ADIPEC 2026
- Co-authored IND-Eval, an industrial time series benchmark paper accepted at a NeurIPS 2026 workshop, and a paper published at ADIPEC 2026
- Received Cognite's 2026 "Master of Agents" Innovation Award for leading adoption of agentic AI development tools across 100+ engineers
- Inventor on 5 patent applications filed with Cognite and 10 automotive patents from Aptiv; received Cognite's 2026 Inventor's Badge

WORK EXPERIENCE

Senior Principal AI Researcher — COGNITE (Oslo, Norway) — 10/2026 - present

Lead Cognite's industrial time series foundation model research, setting research priorities and advancing model performance on benchmarks and industrial use cases.

Principal AI Researcher — COGNITE (Oslo, Norway) — 01/2026 - 09/2026

Established the research lab, from training compute to experimental workflows. Combined hands-on research with leadership of a four-person team to build a state-of-the-art industrial time series foundation model in six months (April–September 2026), using an agent-enhanced software development lifecycle to accelerate iteration.

Selected contributions across both research roles (2026)

- **Synthetic Data**: Implemented synthetic time series generators based on coupled harmonic oscillators for foundation model research
- **Flow Matching**: Defined the research direction and personally designed and evaluated approaches to applying flow matching in time series foundation models
- **Evaluation**: Integrated GIFT-Eval as the primary forecasting benchmark suite; continue extending IND-Eval to evaluate performance on industrial data
- **Training Performance**: Optimized training code through architectural changes, assessed GPU options for model training, and evaluated NorMuon as an alternative to AdamW
- **Research Strategy**: Use literature review, competitor tracking, and controlled experiments to prioritize model improvements against benchmarks and industrial use cases
- **Innovation Leadership**: Serve on Cognite's Patent & Innovation Committee, vetting invention disclosures for novelty and strategic fit and contributing to patent filing decisions

SELECTED RESEARCH PUBLICATIONS

IND-Eval: Aligning Evaluation and Simulation for Industrial Time Series Foundation Models

Vebjørn Haugland, Pavel Gulin, David Foss, **Peet Cremer**, Erlend Aune, Simone Casolo, Andreas Solbrå, Milad Mobarhan. 2026.
Accepted at the Foundation Models for Temporal Systems workshop, NeurIPS 2026; presentation forthcoming, Sydney, December 2026. Industrial forecasting benchmark and study of evaluation-guided synthetic data generation.

Benchmarking Time Series Foundation Models for Oil and Gas Data Forecasting and Well Event Detection

Pavel Gulin, David Foss, Vebjørn Haugland, Erlend Aune, **Peet Cremer**, Simone Casolo.
Published in the ADIPEC 2026 proceedings; paper available on OnePetro.

Further publications in statistical physics and automotive AI: full publication list.

WORK EXPERIENCE (CONTINUED)

Principal Engineer – Atlas AI — COGNITE (Oslo, Norway) — 02/2025 - 12/2025

- **Agent-Enhanced Development**: Led adoption of Cursor, Claude Code, Codex, and Model Context Protocol servers across 100+ engineers, with teams reporting weeks-to-days productivity improvements
- **AI Evaluation**: Designed evaluation frameworks for industrial AI agents and NLP tools interfacing with Cognite's data platform
- **LLM Evaluation**: Benchmarked models from Anthropic, OpenAI, Google, and DeepSeek for industrial knowledge graph applications, informing Atlas AI architecture decisions and publishing an external benchmark report
- **AI Tool Governance & Rollout**: Partnered with Legal, Security, and Procurement to introduce agentic AI tools safely and compliantly, manage costs within budget, and train teams for effective adoption

Senior Director of Engineering — COGNITE (Oslo, Norway) — 02/2024 - 02/2025

- **Hiring & Expansion**: Designed and executed the hiring strategy for Cognite's India Center of Excellence, establishing staffing plans and engineering operations for 20+ roles
- **Engineering Management**: Managed 10–15 developers across data integration, connectivity, and AI initiatives
- **Product Planning**: Contributed to the Contextualization services roadmap, aligning technical architecture with business objectives

Director of Engineering — COGNITE (Oslo, Norway) — 02/2023 - 02/2024

- **Team Leadership**: Expanded responsibility from 1 to 3 teams (up to 18 developers) during management transitions while maintaining delivery commitments and service reliability
- **Team Development**: Enabled support teams to triage customer requests and introduced documented escalation processes, reducing interruptions and engineering team stress
- **Technical Leadership**: Led teams specializing in contextualization of industrial data, data-driven troubleshooting applications, and parsing of engineering diagrams

Senior Machine Learning Engineer and Tech Lead — COGNITE (Oslo, Norway) — 08/2021 - 02/2023

- Led 5 software/ML engineers building production ML services for Cognite's industrial knowledge graph
- Architected and maintained microservices for diagram parsing, named entity recognition, and annotation storage in a SaaS environment
- Built data infrastructure supporting queries over the industrial knowledge graph

AI Lead Developer — APTIV (Wuppertal, Germany) — 12/2020 - 07/2021

- **Radar Perception**: Applied vision-based perception algorithms to automotive radar using LiDAR as a reference sensor; implemented point-cloud autolabeling, training dataset creation, preprocessing, and augmentation
- Led planning and execution of ML and data infrastructure projects for automotive perception tasks
- Contributed to 10 patents across the full Aptiv tenure (2017–2021), covering radar perception training, automotive data storage, and fleet management systems; also co-authored 1 publication (see publication list)

Software Development Expert — APTIV (Wuppertal, Germany) — 07/2017 - 12/2020

- **Data Platform Engineering**: Built end-to-end data platform for automotive sensor data, including recording tools for vehicle sensors, storage infrastructure, labeling systems, and preprocessing workflows
- **ML Inference Infrastructure**: Established microservices for automated inference, enriching incoming sensor data with model predictions
- **Technical Mentorship**: Supervised Master's thesis on GANs for automotive data style transfer

EDUCATION

Ph.D., Theoretical Soft Matter Physics — University of Düsseldorf — 2013 - 2017

Research in statistical physics, numerical simulation, and mathematical modeling, providing a rigorous foundation for experimental design and AI research.

- Dissertation: Mesoscale modeling of magnetic elastomers and gels using finite element methods and density functional theory
- 7 peer-reviewed publications in statistical physics (see publication list)

M.Sc. Physics (German grade: 1.1[1]) — University of Düsseldorf — 2012 - 2013

Focus: Soft Matter, Statistical Physics. Minor: Mathematics. Master's thesis on "Emergent states in active systems" published in a peer-reviewed journal.

B.Sc. Physics (German grade: 1.2[1]) — University of Düsseldorf — 2008 - 2012

Minor: Mathematics. Foundation in mathematical modeling and computational physics. Bachelor's thesis published as a journal article.

Research Internship (DAAD RISE Scholarship) — Yale University, USA — 07/2010 - 10/2010

Awarded a DAAD RISE Worldwide scholarship. Conducted experiments on the optical properties of butterfly wings and developed theoretical models of their optical behavior.

[1] *German grading system: 1.0 (best) to 4.0 (worst)*

SKILLS & LANGUAGES

Programming	Python · C++ · Rust · TypeScript
AI Research	Time Series Foundation Models · Agentic AI · Transformers · Synthetic Data Generation · Flow Matching · Experimental Design · Model Evaluation & Benchmarking
ML Engineering	PyTorch · TensorFlow · scikit-learn · RAG · Model Context Protocol (MCP) · Production ML Systems
Scientific Computing	Statistical Analysis · Numerical Simulation · Mathematical Modeling · NumPy/SciPy
Leadership & Management	Engineering Management · AI Lab Leadership · Technical Hiring · Technical Mentoring · AI Adoption & Governance
Infrastructure & Development	Docker · Kubernetes · Azure · GCP · Linux · Agentic Software Development · Cursor · Claude Code · Codex
Languages	German (native) · English (C1) · Norwegian (B2) · French (A2)

TEACHING & KNOWLEDGE SHARING

Lecturer on Artificial Intelligence — University of Wuppertal, Germany — 10/2020 - 04/2021

Designed and delivered three lectures with exercises on Numerical Optimization (Adam and evolutionary algorithms), Gradient Boosting, and Support Vector Machines.

Co-Organizer, NorwAI 2022 Hackathon — NTNU Trondheim, Norway — 08/2022 - 10/2022

Led technical organization of Data Science hackathon, including dataset curation, task definition, student supervision, and contribution evaluation.

Location & Availability: Based in Norway; open to hybrid roles in Norway and fully remote opportunities.